

# KE-FedRS: Tackling Data Sparsity in Federated Recommendation via Knowledge Enhancement

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## Abstract

Federated recommendation systems (FRSs) have recently gained widespread attention due to their ability to train collaborative recommendation models without exchanging raw user data. However, existing FRSs face a severe challenge of *data sparsity*, which manifests at both the user and item levels. First, user data sparsity: some users may only have a small number of interactions with items, struggling to adequately train the personalized user embedding locally. Second, item data sparsity: some items may only receive a small number of user ratings, causing the global model to lack knowledge about them. Considering these, we propose the **Knowledge Enhanced Federated Recommendation System** named as KE-FedRS, of which the core idea is to enhance the knowledge of users with few interactions and items with few ratings at both the local and global levels. Specifically, at the local level, we introduce an auxiliary user embedding and average and aggregate this auxiliary embedding across similar users, thereby enriching the knowledge of the local user embedding. At the global level, we propose a hybrid client selection strategy based on item embedding discrepancies, prioritizing clients that exhibit greater divergence in item embeddings from others, thus enhancing the knowledge of items with fewer interactions in the global model. We conduct comprehensive experiments on four real-world datasets, and the results show that the proposed method consistently outperforms baseline approaches in terms of HR@10 and NDCG@10.

## CCS Concepts

• **Computing methodologies** → Reasoning about belief and knowledge; Information extraction.

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## Keywords

Federated learning, recommendation system, client selection.

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## 1 Introduction

Federated Recommender Systems (FRSs) [42, 48] combine traditional Recommender Systems [1] with Federated Learning (FL) [17, 39, 40], enabling multiple clients to collaboratively train recommendation models using local data, while keeping each client's data on-device and preventing it from being exposed to the central server or other clients. Due to the protection of user data privacy, FRS has received significant attention in recent years. Specifically, FRS typically treats each user as a client, which locally trains the rating model, user embedding, and item embedding using the user's interaction data with items. To protect the privacy of users, only the rating models and item embeddings are then aggregated on the server [16]. This paradigm can not only provide personalization through local user embeddings but also obtain collaborative generalization knowledge [13] from multiple users by leveraging the global rating model and item embeddings.

In recent years, an increasing number of FRS methods have been proposed [2, 20, 44]. A fundamental approach is to use FedAvg [27] to generate a global model and fine-tune it on the client side [47], thereby training personalized recommendation models for each user. In practical federated recommendation scenarios, item interaction data across clients often exhibits non-IID characteristics due to differences in user preferences. Therefore, to enhance the personalization capability of the rating model, other studies have hypothesized that the overall user population can be partitioned into several clusters with similar preferences [9, 14, 31], with a unique rating model trained for each user cluster. This method identifies similar users by clustering rating model parameters, thereby improving the

adaptability of the rating model when facing heterogeneous client data. Some recent methods have also been proposed to address the cold-start problem in FRS [19, 49], that is, how to generalize the recommendation model to newly joined users, with the core idea being to leverage additional auxiliary information for item recommendation.

Although these methods have achieved significant progress in federated recommendation, they largely overlook the issue of *data sparsity*. In fact, FRSs face a severe challenge of data sparsity at both the user and item levels. *First*, user data is sparse, meaning that some users have only a small number of item interactions, making it difficult for these users with few interactions to adequately train the recommendation model locally. Meanwhile, due to the privacy-preserving requirement for user embeddings, they cannot be aggregated globally, leading to a lack of knowledge in user embeddings. *Second*, item data is also sparse, meaning that certain items are interacted with by only a small number of users, causing the global model to lack knowledge about these items. Specifically, federated recommendation systems typically select a subset of clients uniformly at random, which results in items with fewer user interactions being rarely selected, thereby making it difficult for the global model to learn knowledge about these items.

To address this issue, we propose the Knowledge Enhanced Federated Recommendation System, abbreviated as KE-FedRS, whose core idea is to enhance the knowledge of users with few interactions and items with few ratings at both local and global levels. Specifically, *at the local level*, we introduce an auxiliary user embedding and average and aggregate this auxiliary embedding across similar users, thereby enhancing and supplementing the knowledge of the local user embedding. *At the global level*, we propose a hybrid client selection strategy based on item embedding discrepancies, prioritizing clients that exhibit greater divergence in item embeddings from others—specifically, clients possessing rare items—thus enhancing the global model’s knowledge of items with fewer interactions. We conduct comprehensive experiments on four real-world datasets, and the results show that the proposed method consistently outperforms baseline approaches in terms of HR@10 and NDCG@10. The contributions of this work are:

- We identify the challenge of data sparsity in FRS, where both certain users and items suffer from insufficient interactions, leading to the user embeddings and rating models being unable to learn adequate knowledge.
- We propose a novel method named KE-FedRS, which enhances and supplements knowledge at both the local and global levels through an auxiliary user embedding and an item-embedding-discrepancy-aware client selection strategy, respectively.
- We conduct comprehensive experiments on four benchmark datasets, showing consistent improvements over strong baselines in terms of both HR@10 and NDCG@10.

## 2 Related Work

### 2.1 Clustering Federated Learning

Federated Learning (FL) is a decentralized machine learning framework that allows numerous clients to jointly train a global model

while safeguarding the confidentiality of local data [24, 34]. Depending on the type of participating clients, FL can be categorized into cross-device FL [37], which involves a large number of individual users, and cross-silo FL, where organizations are typically regarded as clients [15]. A wide range of studies has been devoted to addressing key challenges in FL, including communication efficiency [17], privacy preservation [10], data heterogeneity [51], and the cold-start problem [38]. Clustered Federated Learning (CFL) [6, 52] extends FL by clustering similar clients for joint training, thereby improving performance in scenarios with divergent data distributions or non-IID settings and reducing the adverse effects of heterogeneity across clients. To determine cluster partitions, Briggs et al. [6] proposed a hierarchical clustering method that computes the similarity between client models and the global model. Sattler et al. [33] proposed a bipartitioning method utilizing the cosine similarity of client gradients. Mansour et al. [26] allocated each client to the cluster model that minimizes loss. Ruan et al. [32] further observed that each client may also follow a mixture model of multiple data distributions and train both local and cluster models under this setting. Bao et al. [5] apply clustered federated learning to federated recommendation by grouping clients based on learned representations to mitigate data heterogeneity in cross-device FL. However, these approaches have primarily targeted cross-silo FL, whereas research on cross-device CFL remains relatively limited, mainly due to the massive number of client models and the sparsity of local data.

### 2.2 Federal Recommendation

FRS leverage the advantages of FL to safeguard user privacy throughout the recommendation process [43, 45, 46]. FCF [2] was the first framework to incorporate the principle of Collaborative Filtering into FRS. Building upon this, FedMF [8] and FedNCF [30] extended centralized recommendation techniques [13, 18] to federated settings. FedRec [20] investigated explicit feedback in federated recommendation. FedFast [29] improved convergence speed by sampling participating users during each training round. FedeRank [3] enabled users to control the proportion of data to be shared while training personalized factorization models. To further enhance personalization, PFedRec [47] retained the rating prediction module locally and introduced a post-optimization procedure. Furthermore, advanced methodologies such as Graph Neural Networks (GNNs) [23] and meta-learning [21] have been investigated to enhance FRS performance across various subtasks. Many of the above research have employed a server-centric paradigm, in which a single global model is consolidated at the server and subsequently integrated into the clients’ local training. However, when user data distributions exhibit substantial heterogeneity, this strategy may inject noise into the learning process. To mitigate this issue, recent research has shifted toward leveraging clustering methods to construct cluster-specific models, thereby better accommodating diverse user preferences [14]. For instance, PerFedRec [25] clusters similar users based on embedding representations; FPPDM [22] addresses multi-domain recommendation by aligning users through their attributes; and SemiDFEGL [31] implements device-to-device cooperation to improve scalability, allowing users within the same

group to establish local communication graphs for collaborative learning.

### 3 Preliminary and Motivations

**Federated Recommendation.** Let  $U = \{u_i\}_{i=1}^{|U|}$  and  $V = \{v_j\}_{j=1}^{|V|}$  denote the sets of users and items, respectively, where  $|U|$  and  $|V|$  represent their cardinalities. In the context of FRs, each user  $u_i$  is treated as a client and maintains a local dataset  $D_i$ , whose size  $|D_i|$  may vary across users. To preserve user privacy, the local dataset  $D_i$  is never shared with other participants. Each dataset  $D_i$  consists of user–item interaction records  $(u_i, v_j, r_{ij})$ , where  $r_{ij}$  denotes the observed interaction (e.g., rating or feedback) between user  $u_i$  and item  $v_j$ . The global dataset is defined as the union of all local datasets, i.e.,  $D = \{D_1, D_2, \dots, D_{|U|}\}$ . The objective of FRs is to predict the rating  $\hat{y}_{ij}$  of user  $u_i$  on items  $v_j$  that have not been previously interacted with, and then recommend the top- $K$  items with the highest predicted scores. To achieve this, FRS aims to learn a global embedding model  $\theta$ , user embeddings  $E_u = \{e_{u,1}, \dots, e_{u,|U|}\}$ , and item embeddings  $E_v = \{e_{v,1}, \dots, e_{v,|V|}\}$  that minimizes the overall empirical loss on the entire dataset  $D$ .

$$\min_{\theta, E_u, E_v} \mathcal{L}(\theta, E_u, E_v) = \sum_{i=1}^{|U|} \frac{|D_i|}{|D|} \mathcal{L}_i(\theta, e_{u,i}), \quad (1)$$

$$\text{where } \mathcal{L}_i(\theta, e_{u,i}) = - \sum_{j \in |V|} \sum_{k \in |V|} \ln \sigma \left( f(e_{u,i}, e_u^j) - f(e_{u,i}, e_u^k) \right)$$

where  $\mathcal{L}_i(\theta)$  denotes the local loss function of client  $i$ , and  $f(e_{u,i}, e_u^j)$  and  $f(e_{u,i}, e_u^k)$  denote the predicted preference scores of user  $u$  for items  $j$  and  $k$  with the rating model  $f(\cdot)$ , respectively.

Typically, the basic steps of FRs to solve the above loss follows four steps:

- Step 1: The central server initializes a randomly initialized rating model  $\theta$  and distributes it to all clients.
- Step 2: Each client locally training on the received rating model, user embedding, and item embedding using its own dataset.
- Step 3: Then, clients upload their locally updated rating model parameters and item embeddings to the central server.
- Step 4: The central server aggregates the collected local rating models and item embeddings to update the global ones, and then redistributes the updated model and item embeddings to the clients to proceed to the next communication round.

Steps 2-4 constitute a complete communication round. This process is repeated iteratively, with clients and the central server exchanging information in each round until the global model converges.

#### 3.1 Challenges and Motivations

In this section, we investigate the problem of data sparsity.

**Challenge 1: User Data Sparsity.** As shown in Figure 1(a), a large proportion of clients possess only a limited amount of interaction data. Accordingly, the fewer the interactions, the worse the performance with higher RMSE, as shown in Figure 1(b).

**Motivation of Solution:** In FRs, to preserve the privacy of user embeddings, they cannot be aggregated globally like rating model parameters and can only be trained using local item interactions.

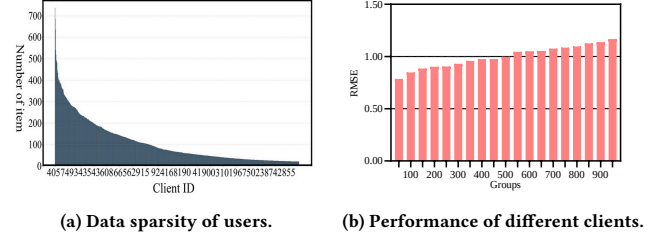


Figure 1: Motivated experiments of user data sparsity on the MovieLens-100K dataset.

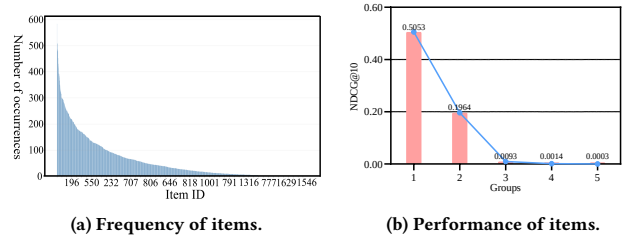


Figure 2: Motivated experiments of item data sparsity on the MovieLens-100K dataset.

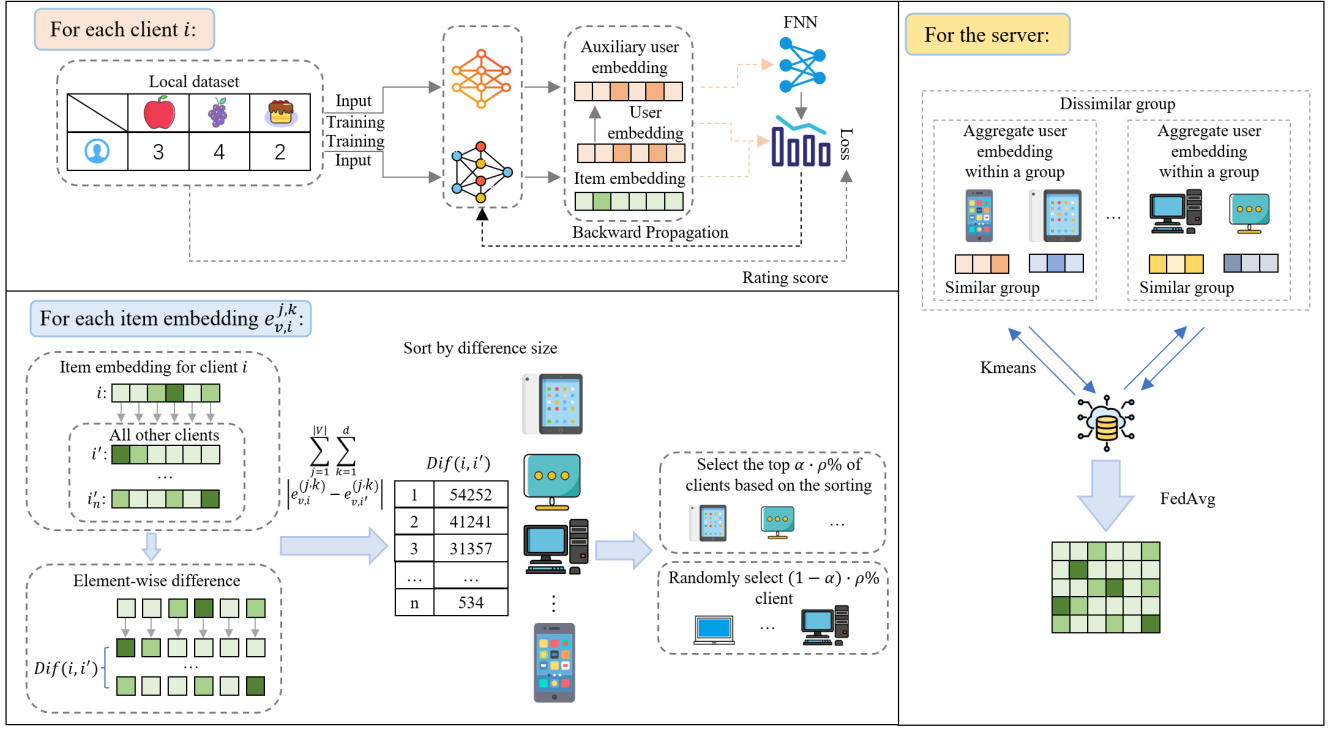
This makes it difficult for clients with sparse interactions to adequately train their user embeddings. To address this issue, a natural question arises: is it possible to enhance the user embedding by leveraging knowledge from other clients while still preserving privacy? In light of this, we propose an auxiliary user embedding approach to supplement knowledge without sharing the original user embeddings.

**Challenge 2: Item Data Sparsity.** As shown in Figure 2(a), a large proportion of items possess only a limited amount of interaction data. Accordingly, the fewer the interactions, the worse the performance, as shown in Figure 2(b).

**Motivation of Solution:** In FRs, federated recommendation systems typically select clients uniformly at random, which results in items with few user interactions being selected only in rare rounds, making it difficult for the global model to learn knowledge about these items. Therefore, a natural idea is: can clients possessing rare items be assigned a higher sampling probability? In light of this, our paper proposes a discrepancy-aware client selection strategy that prioritizes clients whose item embeddings differ significantly from those of other clients.

## 4 Method

In this section, we first provide an overview of the proposed method, including its overall framework and algorithmic workflow. We then present the auxiliary user embedding and the discrepancy-aware client selection strategy approach.



**Figure 3:** At the initial stage of training, the server randomly selects a subset of clients to participate in training. Subsequently, the server computes the discrepancies of item embeddings from each uploaded client and ranks them accordingly. The top  $\alpha \cdot \rho\%$  of clients are selected for the next round of training, while an additional  $(1 - \alpha) \cdot \rho\%$  of clients are randomly chosen to join the process. Each client generates an auxiliary user embedding and uploads it to the central server, where clustering is performed, followed by aggregation within each cluster.

#### 4.1 The Framework and Workflow

Our approach comprises two main stages: (1) grouping clients based on auxiliary user embeddings, and (2) selecting clients for the next training round according to differences in item embeddings. In the client selection stage, the central server compares the item embeddings of each client with those of all other clients, and selects a subset of clients exhibiting the largest differences to participate in the subsequent training round. In the user partitioning stage, we introduce auxiliary user embeddings at each client, after which a portion of the auxiliary user embeddings is uploaded to the central server. The server subsequently categorizes the clients based on these embeddings, and clients perform intra-group aggregation. The process of the proposed framework is depicted in Algorithm 1, whereas Figure 3 depicts the framework.

#### 4.2 Auxiliary User Embedding

Previously, in Section 3, we described the traditional federated recommendation training model, where each client contributes local knowledge by uploading its embedding parameters. Inspired by the FedAvg method—which aggregates client models globally and subsequently allows each client to fine-tune locally on the global model—this framework enables collaborative model building. However, due to substantial differences in user preferences, the

distributions of items across clients may be inconsistent, resulting in potential imbalances. If such heterogeneity in user preferences is not accounted for during global aggregation, it may introduce unwanted noise into the recommendation outcomes.

Moreover, collaborative filtering has been widely demonstrated to be effective in recommender systems, as it leverages ratings or interaction data from neighboring users to capture patterns of similar preferences or behaviors with the target user. Consequently, introducing clustering methods to group clients and items prior to server-side aggregation becomes a natural choice, as it can help uncover and exploit latent patterns or similarities. A typical clustering algorithm is K-Means [28], which partitions data points based on distance metrics. While K-Means demonstrates strong performance in centralized recommendation settings [4, 36, 50], it faces notable challenges in federated recommendation environments.

In federated recommendation, to effectively capture user preference heterogeneity while preserving user privacy, we introduce an auxiliary user embedding, denoted as  $e_u^i$ , which are then uploaded to the central server:

$$e_u^i = [e_u^i(1) || e_u^i(2)] \quad (2)$$

where  $e_u^i(1)$  and  $e_u^i(2)$  represent the first and second halves of the user embedding, respectively, where  $||$  denotes the vector concatenation operation. After the auxiliary user embeddings are uploaded

to the central server, in order to explicitly model user preference heterogeneity prior to cross-client aggregation, we perform clustering on the user embeddings at the server side. Subsequently, representative aggregation is conducted within each cluster. Once all user embeddings are summarized, they form a set.

$$\mathcal{E} = \{e_u^1, e_u^1, \dots, e_u^{|U|}\} \in \mathbb{R}^{|U| \times d} \quad (3)$$

where  $d$  denotes the dimensionality of the user embeddings. At the server side, we perform K-Means clustering on  $E$ , partitioning the users into  $K$  clusters:

$$\min_{\{C_k\}, \{\mu_k\}} \sum_{k=1}^K \sum_{e_i \in C_k} \|e_i - \mu_k\|_2^2, \text{ s.t. } \mu_k = \frac{1}{|C_k|} \sum_{e_i \in C_k} e_i, \quad (4)$$

where  $C_k$  denotes the set of users in cluster  $k$ , and  $\mu_k$  represents its cluster center. After clustering is completed, each user embedding  $e_u^i$  is replaced by the center representation of the cluster to which it belongs:

$$\tilde{e}_u^i = \mu_{c(i)}, c(i) \in \{1, \dots, K\} \quad (5)$$

The replaced representation  $\tilde{e}_u^i$  preserves the inter-user differences, while simultaneously suppressing individual noise through intra-cluster averaging, thereby reducing conflicts during the global aggregation process. Unlike user embeddings, item embeddings  $e_v$  are aggregated using the FedAvg method:

$$\bar{e}_v = \frac{1}{|U|} \sum_i e_v^i \quad (6)$$

Ultimately, the global model constructed by the central server is composed of the user embeddings  $\tilde{e}_u^i$ , which have been averaged within clusters, and the item embeddings  $\bar{e}_v$ , and is distributed to the clients in the next round of training. This approach effectively mitigates the adverse impact of data distribution heterogeneity on federated aggregation while preserving privacy, thereby enhancing the robustness and personalization of the global recommendation model.

### 4.3 Discrepancy-aware Client Selection

In Section 3, we previously described the traditional federated recommendation training framework, in which the server randomly selects a subset of clients to participate in each training round. However, relying solely on random selection may lead to model homogenization, thereby reducing diversity. To address this issue, we propose a hybrid client selection strategy based on project embedding discrepancies. The core idea of this strategy is to prioritize clients that exhibit greater dissimilarity from others, thereby enhancing the diversity and robustness of the global model. To quantify the degree of dissimilarity among clients, we first compute the embedding difference between client  $u_i$  and another client  $u_{i'}$ :

$$Dif(i, i') = \sum_{j=1}^{|V|} \sum_{k=1}^d |e_{v,i}^{(j,k)} - e_{v,i'}^{(j,k)}| \quad (7)$$

where,  $e_{v,i}^{(j,k)}$  denotes the embedding value of item  $v_j$  associated with client  $u_i$  on the  $k$ -th dimension. Subsequently, the overall

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#### Algorithm 1: KE-FedRS: Tackling Data Sparsity in Federated Recommendation via Knowledge Enhancement

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**Input:** Learning rate  $\eta$ ; client participation rate  $\rho$ ; high-difference client ratio  $\alpha$ ; number of clusters  $K$  in K-means.

**Output:** Global model parameters  $\theta$

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**1 Step 1: Initialization.**

2 Randomly initialize the global model  $\theta_g$ .

3 **for**  $r = 1$  **to**  $R$  **do**

4   **(a) Client Selection.**

5   **if**  $r = 1$  **then**

6     Randomly sample  $\rho$  proportion of active clients:  $C_a$ .

7   **else**

8     Each active client trains locally on  $\theta_g$  to obtain  $\theta_i$ ;

9     Compute embedding difference for each client  $i$  by (7);

10    Select top  $\alpha \cdot \rho\%$  clients with largest  $\Delta_i$  and

11     $(1 - \alpha) \cdot \rho\%$  clients randomly to form  $C_a$ .

12   **end**

13   **(b) Local Training.**

14   Each client  $i \in C_a$  updates its local model  $\theta_i$  by minimizing BPR loss.

15   **(c) Aggregation.**

16   - Collect user embeddings  $\{U_i\}$  from  $\{\theta_i\}$ , apply K-means with  $K$  clusters,

17    replace embeddings with corresponding cluster centers, then average across clients. Aggregate item embeddings by mean. - Update global model:  $\theta_g$ .

18   **(d) Model Distribution.**

19   Send updated  $\theta_g$  back to clients for next round training.

20 **end**

21 **Step 2: Evaluation.**

22 Evaluate  $\theta_g$  on the test set using HR@K and NDCG@K.

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dissimilarity of client  $u_i$  is defined as:

$$Dif(i) = \sum_{i' \neq i}^{|U|} Dif(i, i') \quad (8)$$

Based on the heterogeneity measurement results, we first rank all clients in descending order according to their  $Dif(i)$  values, and in each training round, we prioritize the selection of clients with the highest heterogeneity. At the same time, in order to maintain diversity in the training process and to avoid excessive reliance on a small subset of clients, we introduce a stochastic selection mechanism. Specifically, let the overall client participation ratio be denoted as  $\rho$ . A fraction of  $\alpha$  is selected from the set of clients with the highest heterogeneity, while the remaining  $\rho - \alpha$  fraction is randomly sampled from the rest of the clients.

**Table 1: Statistics of datasets.**

| Dataset        | Users | Items  | Interactions | Sparisty |
|----------------|-------|--------|--------------|----------|
| MovieLens-100K | 943   | 1682   | 100,000      | 93.70%   |
| MovieLens-1M   | 6040  | 3706   | 1,000,209    | 95.53%   |
| Lastfm-2K      | 1,600 | 12,454 | 185,650      | 99.07%   |
| FilmTrust      | 1227  | 2059   | 34,888       | 98.62%   |

#### 4.4 Discussion of Privacy

Our methods only uploads auxiliary user embeddings to the central server, with still keeping the user embeddings locally. Unlike directly transmitting complete user embeddings, our approach reduces the transmission of sensitive data, thereby minimizing the risk of privacy leakage. In the context of recommendation systems, when attackers attempt to obtain user representations through gradient inversion or statistical correlation methods, they typically seek to acquire the user’s true ratings, which are computed based on the user and item embeddings obtained by the attackers. Since the attackers cannot access to the user embedding, the true ratings are thus protected.

### 5 Experiments

#### 5.1 Datasets

We conduct experiments on three datasets: Movielens-100k[12], Movielens-1M<sup>1</sup>[12], Lastfm-2K<sup>2</sup> [7] and FilmTrust<sup>3</sup> [11]. We employ a manifold-based leave-one-out data splitting strategy, randomly partitioning the datasets into training, validation, and test sets with an 8:1:1 ratio. We set the learning rate to 0.01 and the number of local training epochs to 5.

#### 5.2 Experimental setup

**5.2.1 Baselines.** We compare the proposed KE-FedRS model with both general and advanced baseline models, including centralized and federated learning approaches. The hyperparameters are set according to the specifications described in the baseline papers.

Cloud-based recommendation methods:

**MF [18]:** MF is a classical recommendation method, whose fundamental idea is to decompose the user–item interaction matrix into the product of low-dimensional user representations and item representations.

**NCF [13]:** NCF employs a multi-layer perceptron to perform a nonlinear combination of user embeddings and item embeddings, thereby enabling the modeling of more complex user–item interactions.

Federated recommendation methods:

**FedMF [8]:** This method uses matrix factorization in a federated setting to reduce information leakage by encrypting user and item embedding gradients.

**FedNCF [30]:** This is the federated adaptation of NCF. It allows local updates of user embeddings on each client while synchronizing item embeddings on the server for global aggregation.

<sup>1</sup><https://grouplens.org/datasets/movielens/>

<sup>2</sup><https://grouplens.org/datasets/hetrec-2011/>

<sup>3</sup><https://guoguibing.github.io/librec/datasets.html>

**FedRecon [35]:** FedRecon combines globally shared representations with local parameter reconstruction, enabling clients to adapt the global model to personalized user preferences.

**SemiDFEGL [31]:** SemiDFEGL adopts a semi-decentralized self-graph learning framework with partial cross-client connections to capture high-order collaboration while preserving privacy in federated recommendation.

**CoFedRec [14]:** CoFedRec employs joint clustering in federated training to aggregate only models from similar users, thereby reducing sparse-feature interference and enhancing personalized recommendation.

**FREIB [41]:** FREIB enhances federated NCF with item bias modeling, global distillation, and prototype alignment to mitigate label-skew-induced bias and improve cross-platform recommendation performance.

**5.2.2 Evaluation metrics.** We evaluate the model performance using Top-K evaluation metrics, including Hit Rate (HR) and Normalized Discounted Cumulative Gain (NDCG). Following the setup from previous work, we fix  $K = 10$  and employ an efficient sampling strategy: for each user, we randomly select 99 unobserved items, and then perform a ranking evaluation over the 100 items, which includes the test items.

#### 5.3 Performance Overview

**Main Results.** This section evaluates the overall performance of the proposed KE-FedRS, with results summarized in Table 2. Several key observations can be drawn. (i) KE-FedRS consistently outperforms all baseline methods across all datasets, with particularly notable gains on the largest dataset, MovieLens-1M. For instance, HR@10 improves from 0.5831 to 0.6775, and NDCG@10 from 0.3216 to 0.3850, highlighting the effectiveness of KE-FedRS in modeling long-tail items and enhancing personalized recommendations. (ii) KE-FedRS achieves the best performance on all four datasets, including both large- and small-scale settings, demonstrating strong robustness and generalization ability. (iii) Compared with centralized approaches, KE-FedRS benefits from selectively involving clients rich in long-tail items and leveraging clustering based on auxiliary user embeddings, which enhances personalization. Moreover, the clustering mechanism effectively aggregates distributed user–item interactions, capturing cross-client user–user and item–item relationships, thereby narrowing the performance gap between federated and centralized recommendation methods.

**Ablation Study.** This section evaluates the contribution of each component in KE-FedRS. The method consists of two key modules: server-side clustering based on auxiliary user embeddings and client selection. In Table 3, we report the corresponding ablated variants as KE-FedRS-w/o cu and KE-FedRS-w/o se. As shown in the results, both components contribute substantially to overall performance. In particular, incorporating client selection yields notable improvements on the MovieLens-1M dataset, highlighting the importance of long-tail items. Meanwhile, server-side clustering effectively groups users with similar preferences, further enhancing personalized recommendation performance.

**Analysis of the Number of User Clusters.** As illustrated in Figure 4, we plot the performance variation of the proposed



**Table 2: Performance on four datasets. (The experimental results are the average of five results.)**

| Methods        |           | Movielens-100K |               | Movielens-1M  |               | LastFM-2K     |               | Film Trust    |               |
|----------------|-----------|----------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                |           | HR@10          | NDCG@10       | HR@10         | NDCG@10       | HR@10         | NDCG@10       | HR@10         | NDCG@10       |
| Centralized    | MF        | 0.1624         | 0.0262        | 0.1058        | 0.0152        | 0.1420        | 0.0435        | 0.3179        | 0.0927        |
|                | NCF       | 0.5814         | 0.3055        | 0.5199        | 0.2848        | 0.1789        | 0.0915        | 0.8138        | 0.6437        |
| Federated      | FedMF     | 0.2486         | 0.1277        | 0.2589        | 0.1235        | 0.2306        | 0.1149        | 0.3755        | 0.1849        |
|                | FedNCF    | 0.4353         | 0.2771        | 0.4236        | 0.2790        | 0.7426        | 0.6614        | 0.8831        | 0.7429        |
|                | FedRecon  | 0.4670         | 0.2963        | 0.4647        | 0.2894        | 0.7762        | 0.6597        | 0.8692        | 0.7404        |
|                | SemiDFEGL | 0.5047         | 0.3080        | 0.5168        | 0.2960        | 0.8058        | 0.6738        | 0.8752        | 0.7909        |
|                | CoFedRec  | 0.5423         | 0.3222        | 0.5396        | 0.3208        | 0.8287        | 0.7019        | 0.8844        | 0.8013        |
|                | FREIB     | 0.5738         | 0.3164        | 0.5683        | 0.3216        | 0.8341        | 0.7175        | 0.9052        | 0.8324        |
| KE-FedRS(Ours) |           | <b>0.6244</b>  | <b>0.3426</b> | <b>0.6775</b> | <b>0.3850</b> | <b>0.8620</b> | <b>0.7452</b> | <b>0.9526</b> | <b>0.8662</b> |

**Table 3: Ablation Study.**

| Method          | Movielens-100K |         | Movielens-1M |         | LastFM-2K |         | Film Trust |         |
|-----------------|----------------|---------|--------------|---------|-----------|---------|------------|---------|
|                 | HR@10          | NDCG@10 | HR@10        | NDCG@10 | HR@10     | NDCG@10 | HR@10      | NDCG@10 |
| KE-FedRS-w/o cu | 0.5793         | 0.3025  | 0.6169       | 0.3418  | 0.8327    | 0.7164  | 0.8947     | 0.8318  |
| KE-FedRS-w/o se | 0.5980         | 0.3223  | 0.6377       | 0.3615  | 0.8479    | 0.7195  | 0.9169     | 0.8412  |
| KE-FedRS        | 0.6244         | 0.3526  | 0.6775       | 0.3850  | 0.8620    | 0.7396  | 0.9526     | 0.8762  |

method under different numbers of user clusters on the MovieLens-100K and MovieLens-1M datasets. The results show that when the number of clusters is too small, the clustering becomes overly coarse. In this case, a large number of users with highly diverse preferences are grouped into only a few large clusters, which fails to capture fine-grained user preferences and consequently reduces the accuracy of personalized recommendations. In contrast, when the number of clusters is excessively large, users are over-partitioned, resulting in clusters that contain too few users. This leads to cluster centroids that lack representativeness and stability, thereby making it difficult to learn generalizable collaborative knowledge.

**Analysis of the Number of Selected Clients.** Performance results are presented in Table 4. We demonstrate the performance variations of the proposed method with respect to changes in the number of selected clients on the MovieLens-100K and MovieLens-1M datasets. It can be observed that when the number of selected clients is too small, the model is unable to adequately explore long-tail items, limiting improvements in personalization and diversity. Conversely, when the number of selected clients is too large, the model fails to effectively learn the popular items, resulting in insufficient gradient updates, which in turn prevents the global model from rapidly learning the fundamental representations of universal user preferences.

**Auxiliary User Embedding Analysis.** Table 5 reports two evaluation metrics under different dimensional settings of user embeddings. The results demonstrate that KE-FedRS achieves its best performance with 16-dimensional user embeddings, indicating that it can effectively preserve user-specific characteristics while mitigating aggregation bias caused by data heterogeneity. When

**Table 4: The number of clients selected for analysis.**

| Percent | Movielens-100K |         | LastFM-2K |         |
|---------|----------------|---------|-----------|---------|
|         | HR@10          | NDCG@10 | HR@10     | NDCG@10 |
| 1%      | 0.6072         | 0.3288  | 0.8334    | 0.6980  |
| 5%      | 0.6105         | 0.3319  | 0.8569    | 0.7166  |
| 10%     | 0.6244         | 0.3526  | 0.8620    | 0.7396  |
| 15%     | 0.6179         | 0.3378  | 0.8497    | 0.7103  |
| 20%     | 0.5962         | 0.3029  | 0.8284    | 0.6838  |

the dimensionality is too low, the embeddings fail to provide sufficiently rich collaborative knowledge for the global model to learn, thereby constraining the accuracy of recommendations. In contrast, when the dimensionality is excessively high, the shared embeddings incorporate an excessive amount of personalized and even noisy features. During the aggregation process, such highly heterogeneous information from different clients tends to conflict, ultimately impairing the model’s generalization ability.

**Client Participation Analysis in Training.** To analyze the dynamic behavior of the proposed hybrid client selection strategy, we examine the client selection process on the MovieLens-100K dataset. As shown in Figure 2b, the selection exhibits clear non-uniformity and temporal dynamics. In the early training stages, client selection is relatively dispersed and partially random, reflecting the stochastic component for exploration. As training proceeds, the selection gradually concentrates on a subset of clients, forming distinct horizontal clusters. This indicates that the strategy effectively identifies and repeatedly selects clients with large model

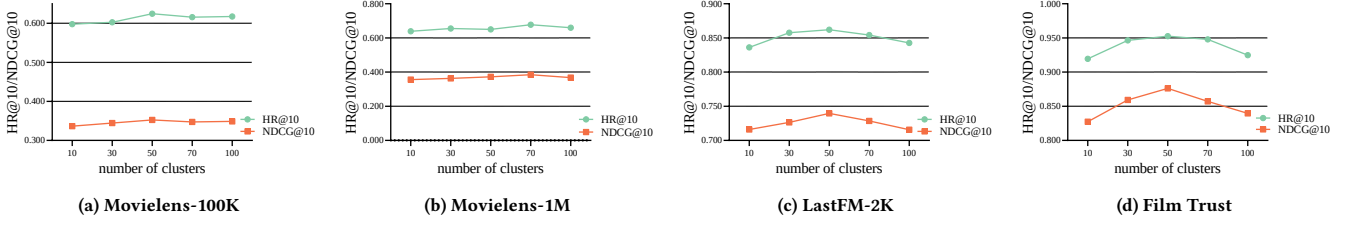


Figure 4: Analysis of the Number of User Clusters.

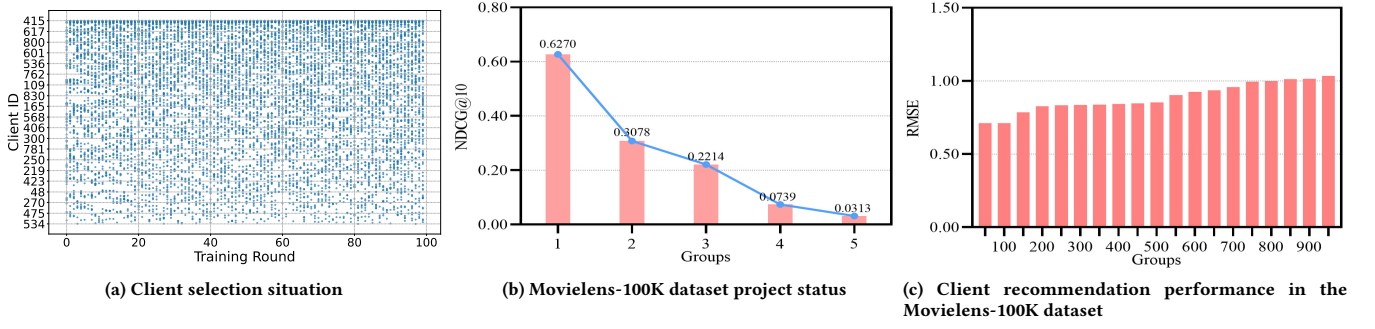


Figure 5: Comparative contrast experiments of motivational experiments.

Table 5: Auxiliary user embedding scale analysis

| Size | Movielens-100K |         | Movielens-1M |         |
|------|----------------|---------|--------------|---------|
|      | HR@10          | NDCG@10 | HR@10        | NDCG@10 |
| 8    | 0.5470         | 0.2877  | 0.5942       | 0.3236  |
| 16   | 0.6244         | 0.3526  | 0.6775       | 0.3850  |
| 32   | 0.6031         | 0.3346  | 0.6683       | 0.3826  |
| 64   | 0.6015         | 0.3315  | 0.6561       | 0.3777  |

discrepancies and a higher proportion of long-tail items, validating its effectiveness.

**Analysis of Items with Different Frequencies.** To examine the effect of item popularity on federated recommendation performance, we group items into ten popularity levels based on their interaction frequency in the training data and evaluate the average NDCG@10 for each group. As shown in Figure 5b, KE-FedRS consistently outperforms FedNCF across all popularity levels, demonstrating its superior recommendation capability for both popular and long-tail items.

**Performance Analysis under Different Levels of User Data Sparsity.** Clients are ranked by the number of interacted items and partitioned into ten groups, with RMSE used as the evaluation metric. As shown in Figure 5c, KE-FedRS consistently outperforms FedNCF across all groups, especially for clients with fewer interactions. This demonstrates that KE-FedRS generalizes more effectively

under data sparsity and learns more robust user preference representations.

## 6 Conclusion

This paper addresses the challenge of user and item data sparsity in existing federated recommendation systems. To tackle this challenge, we propose KE-FedRS, a novel framework designed to enrich the representation of users with limited interaction history and items with sparse ratings, operating at both local and global scales. At the local scale, we design an auxiliary user embedding module that captures collaborative signals from users with similar behavior patterns. By averaging and aggregating this auxiliary embedding across neighboring users, we effectively augment the representational capacity of the local user embedding. At the global scale, we develop a discrepancy-aware client selection mechanism that identifies and prioritizes participants whose item embeddings significantly deviate from the global consensus—typically those interacting with less common (rare) items. This strategy ensures that underrepresented items receive adequate attention during model aggregation, thereby improving the global model’s understanding of rare items. We conduct comprehensive experimental evaluations on four real-world datasets. The experimental results demonstrate that the proposed method significantly outperforms existing baseline models in terms of both HR@10 and NDCG@10.

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